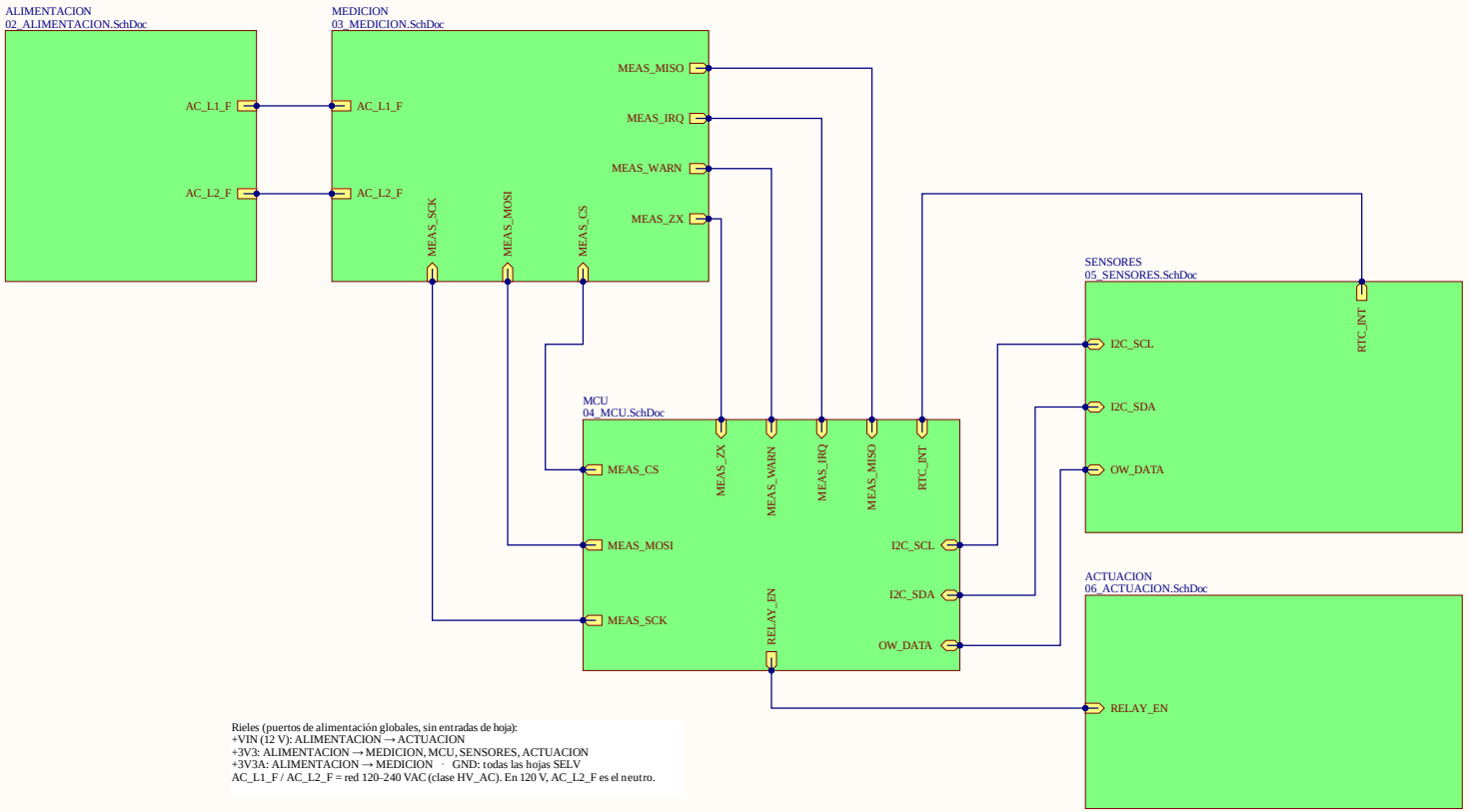
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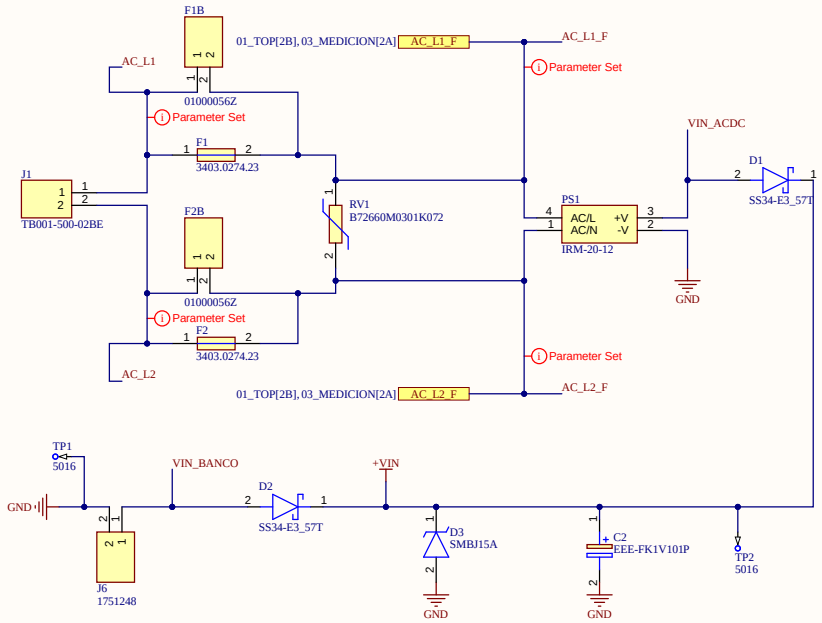


Rieles (puertos de alimentación globales, sin entradas de hoja):
+VIN (12 V): ALIMENTACION → ACTUACION
+3V3: ALIMENTACION → MEDICION, MCU, SENSORES, ACTUACION
+3V3A: ALIMENTACION → MEDICION · GND: todas las hojas SELV
AC_L1_F / AC_L2_F = red 120-240 VAC (clase HV_AC). En 120 V, AC_L2_F es el neutro.

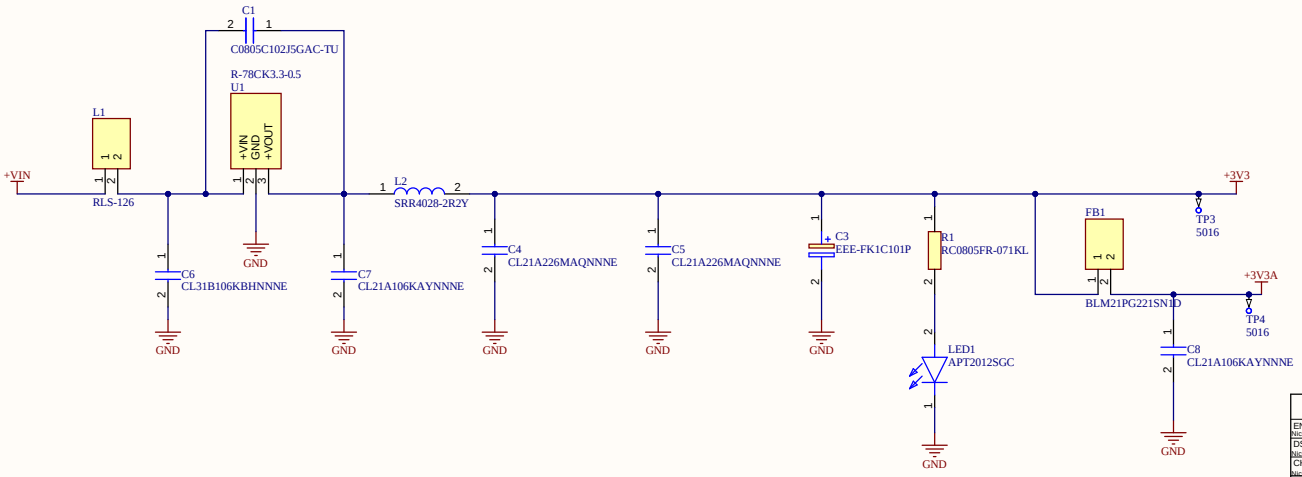
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DSN: Nicolás Gómez Ramírez	10/12/2026	PROJECT REVISION: 0.1	DOCUMENT REVISION: Not in version control
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PCB DWG:		SCALE:	FILE NAME 01_TOP.SchDoc
			SHEET 1 OF 6

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Alimentar desde breaker exclusivo de 2-6 A (bipolar en 240 V). J1: 1 = L1/L, 2 = L2/N. J6: 12 V de banco (9–15 V).



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Parameter Set

HV_D1 HV_D2 HV_D3

01_TOP[2B], 02_ALIMENTACION[3A] AC_L1_F — R2 ERJ-P08F1004V — R3 ERJ-P08F1004V — R4 ERJ-P08F1004V — R5 ERJ-P08F1004V — HV_INP

R6 ERA-6AEB103V — C9 C0805C102J5GAC-TU

01_TOP[2B], 02_ALIMENTACION[3B] AC_L2_F — AC_L2_F

U2 AMC3330DWE

HV_DCDC 1 — DCDC_OUT 16 AMC_LDO
AC_L2_F 2 — DCDC_HGND 15 DCDC_GND — GND
HV_DCDC 3 — HLDO_IN 14 DIAG — AMC_DIAG
AC_L2_F 4 — NC 13 LDO_OUT — AMC_LDO — +3V3
HV_LDO 5 — HLDO_OUT 12 VDD
HV_INP 6 — INP 11 OUTP — AMC_OUTP
AC_L2_F 7 — INN 10 OUTN — AMC_OUTN
AC_L2_F 8 — HGND 9 GND — GND

HV_DCDC

C10 CL21B105KAFNNNE — C11 C0805C102J5GAC-TU — C12 CL21B104KBCNNNC — AC_L2_F

HV_LDO

C16 C3216NP01H104J160AA — C17 C0805C102J5GAC-TU — AC_L2_F

AMC_LDO — C13 CL21B104KBCNNNC — GND

+3V3 — C14 CL21B105KAFNNNE — C15 C0805C102J5GAC-TU — R7 RC0805FR-0747KL — AMC_DIAG — TP5 5016

AMC_OUTP — C18 CL21A106KAYNNNE — R9 ERA-6AEB203V — ATM_VP

R10 ERA-6AEB133V — C19 C0805C472J5GACTU

AMC_OUTN — C21 CL21A106KAYNNNE — R12 ERA-6AEB203V — ATM_VN

R14 ERA-6AEB133V — C22 C0805C472J5GACTU

Parameter Set

CT_P — R8 RC0805FR-071KL

J3 TIP_SWITCH SJ1-3515-SMT-TR — TIP — RING_SWITCH — RING — SLEEVE

D4 SMAJ5.0CA

R11 ERA-6AHD100V — ATM_IP — C20 C1206C333J5GACTU

R13 ERA-6AHD100V

CT_N — R15 RC0805FR-071KL — ATM_IN — C23 C1206C333J5GACTU

ATM_CF2 — TP6 5016

U3 ATM90E26-YU-R

MMD1 1 — MMD0 28 +3V3
DGND 2 — SDI/URX 27 MEAS_MOSI — 01_TOP[3B], 04_MCU[1B]
DVDD 3 — SDO/UTX 26 MEAS_MISO — 01_TOP[3A], 04_MCU[1B]
RESET 4 — SCLK 25 MEAS_SCK — 01_TOP[3B], 04_MCU[1B]
AVDD 5 — CS 24 MEAS_CS
AGND_1 6 — OSCO 23
I2P 7 — OSCI 22 XTAL_O
I2N 8 — ZX 21 MEAS_ZX — 01_TOP[3B], 04_MCU[1B] — XTAL_I
RESV_LOW 9 — IRQ 20 MEAS_IRQ — 01_TOP[3B], 04_MCU[1B]
I1P 10 — CF2 19 ATM_CF2
I1N 11 — CF1 18 ATM_CF1
USEL 12 — WARNOUT 17 MEAS_WARN — 01_TOP[3B], 04_MCU[1B]
VREF 13 — VP 16 ATM_VP
AGND_2 14 — VN 15 ATM_VN

+3V3 — C24 CL21A106KAYNNNE — C25 CL21B104KBCNNNC — GND

ATM_VREF — C30 CL21B105KAFNNNE — C31 C0805C102J5GAC-TU — GND

+3V3A — C28 CL21B104KBCNNNC — GND

+3V3 — R16 RC0805FR-0710KL — ATM_RST — C29 CL21B104KBCNNNC — GND

XTAL_I — Y1 ECS-81.92-12-30Q-EN-TR — XTAL_O — C26 C0805C120J5GACTU — C27 C0805C120J5GACTU — GND

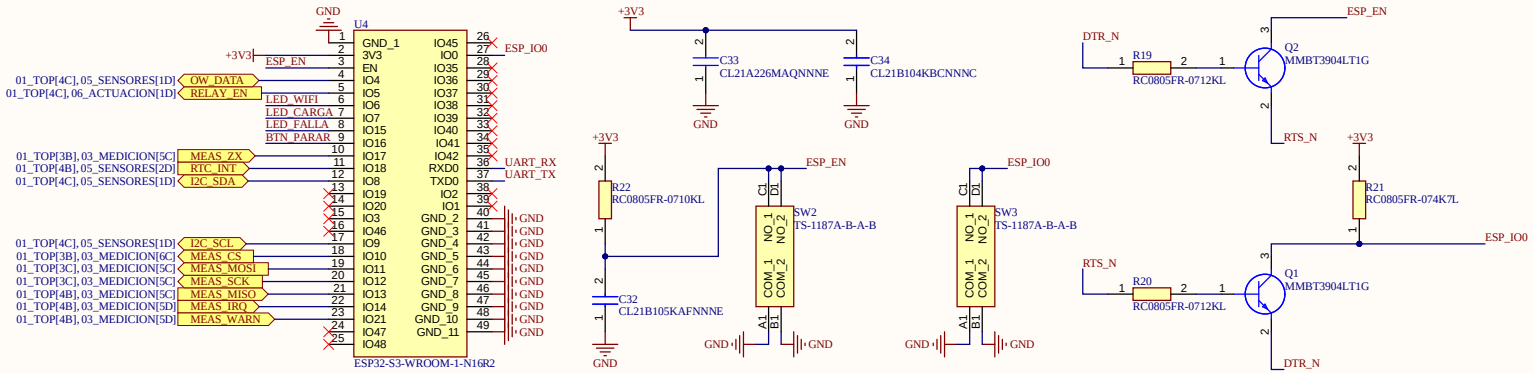
ATM_CF1 — R17 RC0805FR-071KL — LED2 APT2012SGC — GND

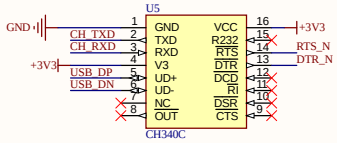
+3V3 — R18 RC0805FR-0710KL — MEAS_CS

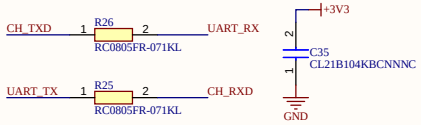
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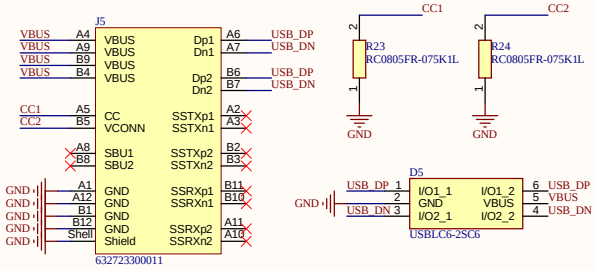
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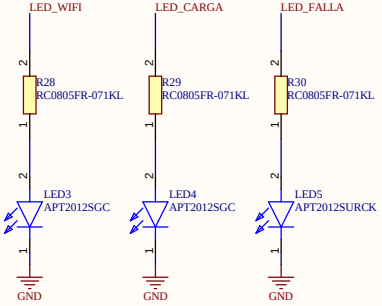
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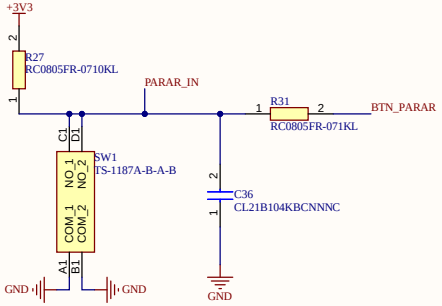


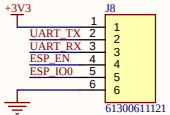


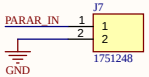












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01_TOP[4B], 04_MCU[1B] I2C_SCL

01_TOP[4B], 04_MCU[1B] I2C_SDA

01_TOP[4B], 04_MCU[1A] OW_DATA

01_TOP[5B], 04_MCU[1B] RTC_INT

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PROJECT: Control de Carga

PROJECT REVISION: 0.1

DOCUMENT REVISION: Not in version control

DESIGN BY: Nicolás Gómez Ramírez

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01_TOP[4C], 04_MCU[1A] RELAY_EN

APPROVALS	DATE
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PROJECT: Control de Carga

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DOCUMENT REVISION: Not in version control

DESIGN BY: Nicolás Gómez Ramírez

TITLE: *

SIZE	CAGE CODE	DWG NO.	REV
A3			

SCALE: | FILE NAME: 06_ACTUACION.SchDoc | SHEET 6 OF 6

Comment	Description	Designator	Footprint	LibRef	Quantity
C0805C102J5GAC-TU	Capacitor	C1, C9, C11, C15, C17, C31	CAPC2012X88N	C0805C102J5GAC-TU	6
EEE-FK1V101P	Capacitor Polarised	C2	EEHZT1E331UP	EEE-FK1V101P	1
EEE-FK1C101P	Capacitor Polarised	C3	EEE1AA101SP	EEE-FK1C101P	1
CL21A226MAQNNNE	Capacitor	C4, C5, C33	CAPC2012X140N	CL21A226MAQNNNE	3
CL31B106KBHNNNE	Capacitor	C6	CAPC3216X180N	CL31B106KBHNNNE	1
CL21A106KAYNNNE	Capacitor	C7, C8, C18, C21, C24	CAPC2012X145N	CL21A106KAYNNNE	5
CL21B105KAFNNNE	Capacitor	C10, C14, C30, C32	CAPC2012X88N	CL21B105KAFNNNE	4
CL21B104KBCNNNC	Capacitor	C12, C13, C25, C28, C29, C34, C35, C36	CAPC2012X88N	CL21B104KBCNNNC	8
C3216NP01H104J160AA	Capacitor	C16	CAPC3216X180N	C3216NP01H104J160AA	1
C0805C472J5GACTU	Capacitor	C19, C22	CAPC2012X88N	C0805C472J5GACTU	2
C1206C333J5GACTU	Capacitor	C20, C23	CAPC3216X180N	C1206C333J5GACTU	2
C0805C120J5GACTU	Capacitor	C26, C27	CAPC2012X88N	C0805C120J5GACTU	2
SS34-E3_57T	Schottky Diode	D1, D2	DIOM7959X262N	SS34-E3_57T	2
SMBJ15A	TVS Diode (Uni-directional)	D3	DIOM5436X244N	SMBJ15A	1
SMAJ5.0CA	TVS Diode (Bi-directional)	D4	DIONM5026X229N	SMAJ5.0CA	1
USBLC6-2SC6	Integrated Circuit	D5	SOT95P280X145-6N	USBLC6-2SC6	1
3403.0274.23	Fuse	F1, F2	3403027423	3403.0274.23	2
01000056Z	Undefined or Miscellaneous	F1B, F2B	01000056Z	01000056Z	2
BLM21PG221SN1D	Ferrite Bead	FB1	BEADC2012X105N	BLM21PG221SN1D	1
TB001-500-02BE	Connector	J1	TB001-500-02BE	TB001-500-02BE	1
SJ1-3515-SMT-TR	Connector	J3	SJ13515SMTTR	SJ1-3515-SMT-TR	1
632723300011	Conn USB 3.1 Type C RCP 24 POS 0.5mm/0.4mm Solder RA SMD/Thru-Hole 24 Terminal 1 Port T/R	J5	632723300011	CMP-1502-03214-4	1
1751248	Connector	J6, J7	1751248	1751248	2
61300611121	Connector	J8	HDRV6W79P0X254_1X6_1524X254X894P	61300611121	1
RLS-126	Power Supply	L1	RLS126	RLS-126	1
SRR4028-2R2Y	Inductor	L2	SRR4028	SRR4028-2R2Y	1
APT2012SGC	LED	LED1, LED2, LED3, LED4	LEDC2012X85N	APT2012SGC	4
APT2012SURCK	LED	LED5	LEDC2012X85N	APT2012SURCK	1
IRM-20-12	Power Supply	PS1	IRM2012	IRM-20-12	1
MMBT3904LT1G	Transistor BJT NPN	Q1, Q2	SOT96P237X111-3N	MMBT3904LT1G	2
RC0805FR-071KL	Resistor	R1, R8, R15, R17, R25, R26, R28, R29, R30, R31	RESC2012X60N	RC0805FR-071KL	10
ERJ-P08F1004V	Resistor	R2, R3, R4, R5	ERJP08_(1206)	ERJ-P08F1004V	4
ERA-6AEB103V	Resistor	R6	RESC2012X60N	ERA-6AEB103V	1
RC0805FR-0747KL	Resistor	R7	RESC2012X60N	RC0805FR-0747KL	1
ERA-6AEB203V	Resistor	R9, R12	RESC2012X60N	ERA-6AEB203V	2
ERA-6AEB133V	Resistor	R10, R14	RESC2012X60N	ERA-6AEB133V	2
ERA-6AHD100V	Resistor	R11, R13	RESC2012X60N	ERA-6AHD100V	2
RC0805FR-0710KL	Resistor	R16, R18, R22, R27	RESC2012X60N	RC0805FR-0710KL	4
RC0805FR-0712KL	Resistor	R19, R20	RESC2012X60N	RC0805FR-0712KL	2
RC0805FR-074K7L	Resistor	R21	RESC2012X60N	RC0805FR-074K7L	1
RC0805FR-075K1L	Resistor	R23, R24	RESC2012X60N	RC0805FR-075K1L	2
B72660M0301K072	Varistor	RV1	CU4032	B72660M0301K072	1
TS-1187A-B-A-B	Tact switch 5.1 x 5.1 x 1.5 mm, 160 gf, 50 mA 12 VDC	SW1, SW2, SW3	TS1187ABAB	TS-1187A-B-A-B	3
5016	Test Point, 1-Pin SMT, RoHS, Tape and Reel	TP1, TP2, TP3, TP4, TP5, TP6	KEYSTONE-5016_V	CMP-2000-05178-1	6
R-78CK3.3-0.5	Regulador conmutado 3.3 V 0.5 A, SIP3	U1	R78CK3305	R-78CK3.3-0.5	1
AMC3330DWE	Integrated Circuit	U2	SOIC127P1030X265-16N	AMC3330DWE	1
ATM90E26-YU-R	Integrated Circuit	U3	SOP65P778X199-28N	ATM90E26-YU-R	1
ESP32-S3-WROOM-1-N16R2	Integrated Circuit	U4	ESP32S3WROOM1N16R2	ESP32-S3-WROOM-1-N16R2	1
CH340C	USB to UART bridge	U5	SOIC127P600X175-16N	CH340C	1
ECS-81.92-12-30Q-EN-TR	Crystal or Oscillator	Y1	ECS81921230QENTR	ECS-81.92-12-30Q-EN-TR	1